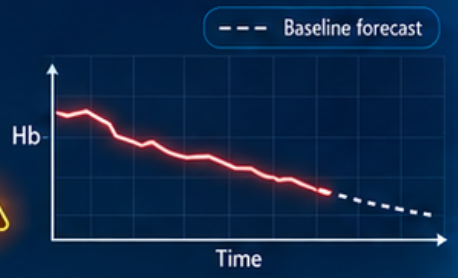
# CKD-Related Clinical Risks & Predictive Intelligence

## Sri Lankan Longitudinal Renal & Hemodialysis Data

### Anemia / Hb Deterioration

Monitors declining Hemoglobin
Predicts next-period Hb risk



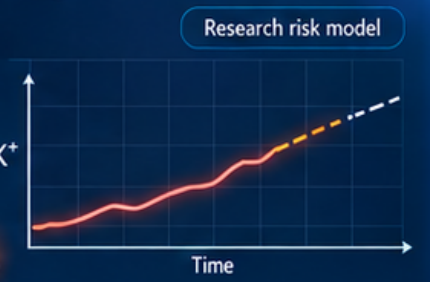



Hb | Albumin | Urea | Creatinine | Calcium | Potassium

### Hyperkalemia Monitoring

Detects rising Potassium patterns
Supports early clinical review



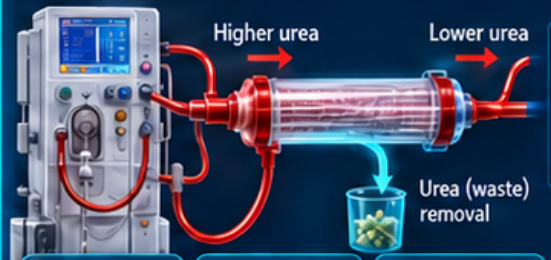




Pre-HD K+ | Post-HD K+ | K+ Difference | Creatinine | Longitudinal K+ Trend

### Dialysis Adequacy / URR

Measures effectiveness of urea removal
Forecasts next-period inadequate dialysis risk

Research forecast




$$URR = \left(\frac{\text{Pre-HD Urea} - \text{Post-HD Urea}}{\text{Pre-HD Urea}}\right) \times 100$$

Pre-HD Urea | Post-HD Urea | Current JRR | Previous URR | URR Trend



**Chronic Kidney Disease / Kidney Failure (G5D)**
Longitudinal Clinical Monitoring

G5D: verified maintenance dialysis

### Renal Clinical Deterioration

Combines multiple renal trends
Stable • Improving • Declining

Trend composite




Creatinine ↑ | Hemoglobin ↓
Urea ↑ | Albumin ↓
Potassium ↑ | URR ↓

Sri Lankan Longitudinal Patient Data → Clinical Data Validation → Predictive Intelligence → Explainable AI → Clinician Review

Longitudinal Data | Explainable AI | Clinician-in-the-Loop | FHIR R4 Compatible

Research concept • Illustrative trends • Clinical review signals